Project Report: Analyze Voter Turnout

Uyen Nguyen

December 18, 2025

1 Introduction

Voter complacency is the idea that when voters believe an election outcome is already decided, or when they feel their side is safe, they become less motivated to show up. In other words, as perceived uncertainty drops, the incentive to participate drops as well. It is a theory about how confidence and expectations can quietly reduce turnout and engagement.

In this project, we test whether evidence consistent with voter complacency appears within each U.S. presidential election cycle we study (2008, 2016, and 2020). Using county-level election results, we focus on the relationship between primary-election dynamics and general-election participation within the same year. The intuition is straightforward: the primary season reveals how competitive or uncertain a party's race is, and that uncertainty may shape voter enthusiasm and turnout once the election transitions to the general phase.

We begin by standardizing county-level primary and general-election data across all three cycles. We then perform a weighted, multi-state analysis of Republican general-election vote-share overperformance relative to Republican primary vote share, allowing us to compare how primary strength translates into general outcomes across states of different sizes. To more directly test for complacency, we introduce a general-election multiplier model that compares general-election turnout to primary turnout at the county level, capturing how participation expands (or fails to expand) from the primary to the general election. Finally, we compute Shannon entropy for both Democratic and Republican primaries and analyze it as a time-series measure of electoral uncertainty over time. Taken together, these analyses provide multiple lenses through which to examine whether lower primary-level uncertainty is associated with patterns consistent with voter complacency in the general election.

2 Data Cleaning

This project analyzes three U.S. presidential election cycles (2008, 2016, and 2020) using election returns from the OpenElections repository [GitHub]. The goal of the data cleaning stage is to transform state-by-state CSV files into a single standardized format that supports cross-state and cross-year analysis.

In particular, we aim for every cleaned dataset to:

1. Clearly separate primary versus general election results
2. Separate Democratic and Republican contests
3. Preserve candidate-level vote totals when available
4. Include some aggregate totals for modeling and comparison.

Across all years, the workflow follows the same high-level structure: read in raw state files, normalize column names and county/precinct fields, filter to presidential rows, extract vote counts, reshape to a wide format with consistent prefixes (e.g., `pri_dem_*`, `pri_rep_*`, `gen_*`), and compute totals such as `dem_pri_total`, `rep_pri_total`, and `gen_total`. Cleaned outputs are saved as one CSV per state.

2.1 2008 Cycle

The 2008 cleaning is implemented as one notebook per state. Each notebook loads raw files from data/raw/2008/<STATE>/, normalizes headers (lowercasing, trimming whitespace, and standardizing common aliases), and identifies a stable geographic key, preferring counties when precinct-level reporting is inconsistent or incomplete. We then filter to presidential rows only for simplicity.

Steps are repeated for both the primary and general. Candidate names are lightly cleaned for consistency, vote columns are coerced to numeric, and duplicate reporting units are aggregated when necessary. Finally, the data are pivoted to wide format so each geography corresponds to a single row, and total vote columns (`dem_pri_total`, `rep_pri_total`, `gen_total`) are computed. Each cleaned state file is written to data/processed/2008/<STATE>.csv and used directly in downstream EDA and modeling.

After cleaning, the following states have complete data or less than 5% missing counties: Arizona (AZ), Delaware (DE), Georgia (GA), Idaho (ID), Illinois (IL), Indiana (IN), Louisiana (LA), Massachusetts (MA), Montana (MT), Ohio (OH), Oregon (OR), Pennsylvania (PA), Tennessee (TN), Texas (TX), and Wisconsin (WI). Together, these states represent 30% of the total 50 states.

2.2 2016 Cycle

For the 2016 cycle, I use an existing set of cleaned CSV files produced during prior summer research (VoterTurnout). These data are already standardized and analysis-ready, but they are currently reported at the precinct level. Since the rest of the project operates at the county level, additional aggregation steps are performed during exploratory data analysis to bring the 2016 data into alignment with the other cycles.

2.3 2020 Cycle

For the 2020 presidential election cycle, I built a single automated ETL pipeline (`cleaning_pipeline.py`) to standardize multiple states in one run. Starting from the OpenElections GitHub repositories, I manually selected states with sufficiently complete presidential primary and general-election data and placed the raw CSVs under `data/raw/2020/<STATE>/`. The pipeline then performs the remaining extraction, transformation, and loading steps automatically.

The pipeline loops over each state folder, identifies primary and general-election files based on filename patterns, and concatenates multiple files per stage when necessary. It includes basic robustness checks to handle malformed CSVs so that individual file issues do not halt the entire run.

The transformation stage performs most of the substantive work. Column names are normalized, and a minimal schema is enforced so that presidential contests can be reliably identified. The pipeline constructs a single county-level geography key by searching across common aliases and renaming the first valid match to `county`. This choice enforces uniformity across states even when counties are not the native reporting unit, for example, parishes in Louisiana, boroughs or census areas in Alaska, and town-based reporting in states like Connecticut. These units are treated as county-equivalents so that downstream analysis can operate on a consistent geographic dimension.

Presidential contest rows are isolated by filtering the `office` field, and administrative summary rows (e.g., statewide totals) are explicitly removed to prevent double counting. Vote totals are then extracted using a tiered strategy that prioritizes explicit `votes` columns when available and falls back to validated numeric fields when necessary. Party labels are normalized into a compact set of codes, candidate names are cleaned into column-safe identifiers, and county names are standardized to reduce artificial duplicates.

The pipeline aggregates all data to the county level and reshapes the result into wide format, producing one row per county with stage-specific vote columns. Primary and general datasets are merged on `county`, and aggregate totals (`gen_total`, `rep_pri_total`, `dem_pri_total`) are computed. Each state's cleaned output is written to `data/processed/2020/<STATE>.csv`, along with a run summary used to assess data completeness.

After running the pipeline, 19 states meet the completeness criteria (or have less than 5% missingness) and are retained for analysis. These states are: Alabama (AL), Arizona (AR), Colorado (CO), Connecticut (CT), Delaware (DE), Georgia (GA), Idaho (ID), Indiana (IN), Louisiana (LA), Massachusetts (MA), Mississippi (MS), New Hampshire (NH), New York (NY), Oregon (OR), South Carolina (SC), South Dakota (SD), Washington (WA), West Virginia (WV), and Wisconsin (WI).

However, there are several limitations of the 2020 pipeline. The script is intentionally cycle-specific, including filename patterns, party normalization rules, and candidate-naming conventions chosen to align with the 2008 and 2016 data. In a few cases, I introduced state-specific fixes (for example, a manual town-to-county mapping for Massachusetts), which improve usability but may introduce minor transcription risk.

Additionally, while the pipeline explicitly removes statewide "Totals" rows and other non-geographic summaries, it does not fully resolve all edge cases. Columns such as `WRITEIN`, `OTHERS`, or similar categories may be inconsistently treated across states, and the fallback vote-detection logic prioritizes robustness over perfect precision. Thus, the pipeline should be viewed as a strong baseline rather than a universal cleaner. Extending it to new cycles or additional states would require updating hard-coded assumptions and adding validation checks.

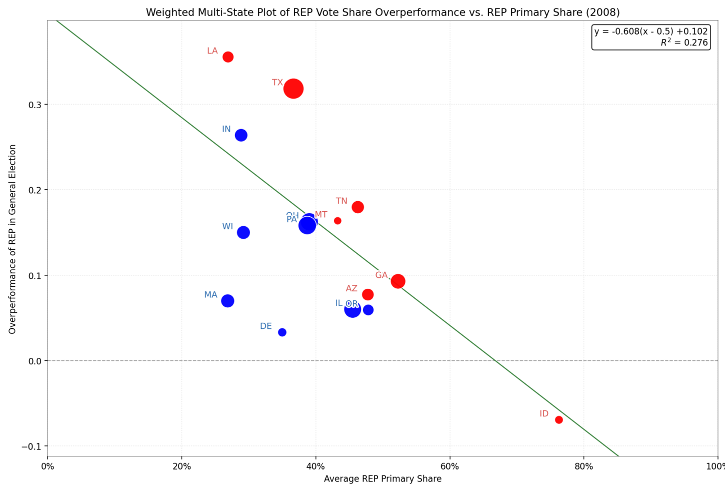
3 Exploratory Data Analysis

Before introducing formal models, we begin with exploratory data analysis to understand how primary-election vote shares translate into general-election outcomes. For each election cycle and each state, we reduce the cleaned data to four core quantities: `rep_primary_total`, `dem_primary_total`, `rep_general_total`, and `dem_general_total`.

From these, we compute Republican vote share in both the primary and general election, as well as total two-party turnout in each stage. The key outcome variable in this section is Republican vote-share overperformance, defined as

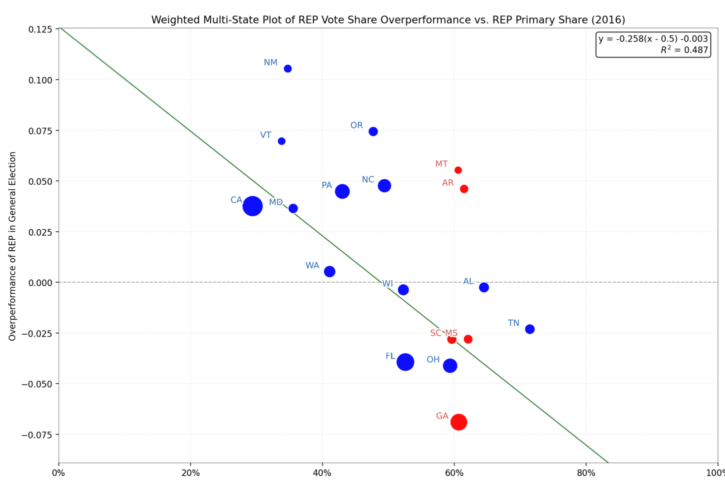
$$\text{REP overperformance} = \text{REP general share} - \text{REP primary share}$$

We then construct a weighted multi-state scatter plot for each election year. Each point represents a state, positioned by its average Republican primary share (x-axis) and Republican overperformance in the general election (y-axis). Point size is proportional to total general-election turnout. A fitted linear trend line then summarizes the overall relationship across states.

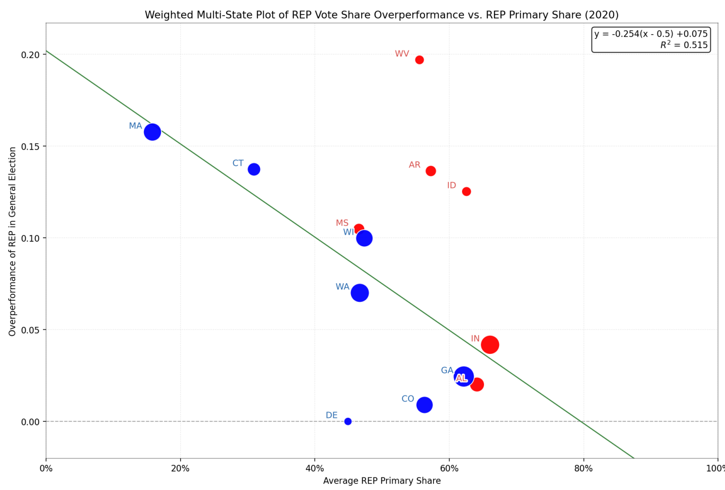


For the 2008 election cycle, the weighted multi-state plot reveals a clear negative relationship between Republican primary share and Republican general-election overperformance. States where Republicans performed strongly in the primary tend to show smaller gains, or even losses, in their general-election vote share relative to the primary. Conversely, states with weaker Republican primary performance often exhibit positive overperformance in the general election.

This pattern is suggest the voter complacency: in states where the Republican primary is already strong, there is less room (or less incentive) for further expansion in the general election, while weaker primary states may experience consolidation or crossover support in November. The moderate R^2 suggests that primary strength explains some but not all variation.



The 2016 cycle exhibits a similar but more tightly clustered pattern. The negative slope persists, indicating that higher Republican primary share is again associated with lower Republican overperformance in the general election. However, compared to 2008, the points lie closer to the fitted line, and the overall fit improves.



In 2020, the negative relationship between Republican primary share and general-election overperformance becomes even more pronounced. The fitted line is steeper, and the explanatory power of the linear trend is higher than in the previous cycles. States with very high Republican primary shares tend to exhibit little to no overperformance in the general election, while states with lower primary shares are more likely to show positive gains.

This cycle stands out both because of its stronger linear pattern and because of the institutional context. The 2020 primaries were shaped by incumbency, uneven contestation, and pandemic-era disruptions, all of which may amplify the contrast between primary dominance and general-election dynamics. While this makes causal interpretation difficult, the descriptive pattern is striking: primary strength appears increasingly decoupled from general-election gains as Republican primary share rises.

However, I note that out of the 19 states with cleaned data files, only 15 are used in these plots. Louisiana, New Hampshire, New York, South Carolina, and South Dakota are excluded due to irregularities in their total-vote columns that require further investigation. These issues prevented those states from being reliably included in the figures above.

4 Complacency: General-Election Multipliers

To test the voter complacency, we develop the following descriptive model. For each state and election year, we compute two-party (DEM vs. REP) turnout totals for both the primary and the general election. For each party, we define a general-election multiplier as

$$r_{\text{party}} = \frac{G_{\text{party}}}{P_{\text{party}}}$$

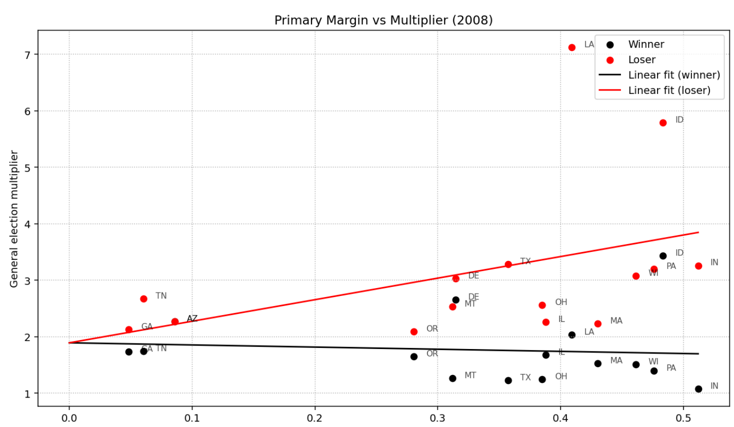
where P_{party} denotes the total number of votes cast for that party in the primary and G_{party} denotes that number in the general. Intuitively, this multiplier captures how much a party's electorate expands (or contracts) when moving from the primary to the general election.

We then relabel parties within each state–year as the primary winner or primary loser. The primary winner is defined as the party with higher two-party primary turnout in that state–year, and the primary loser is the other. Under this relabeling, each state–year yields two multipliers,

$$r^W = \frac{G^W}{P^W}, \qquad r^L = \frac{G^L}{P^L}$$

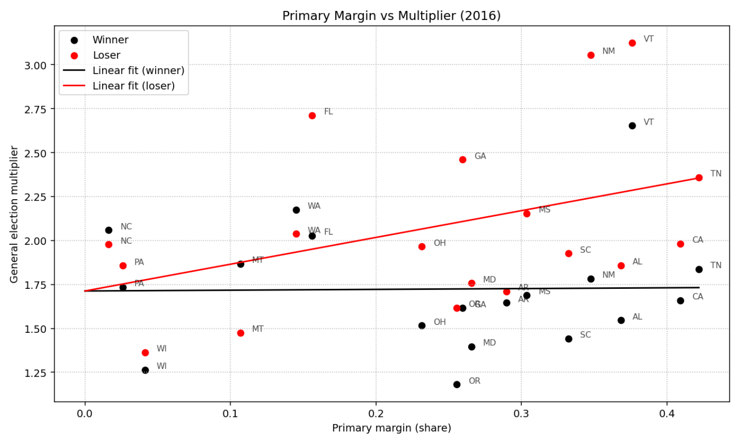
where superscripts W and L denote the primary winner and loser, respectively.

We now plot r^W and r^L against the primary margin and fit separate linear trend lines for winners and losers. We also fix a single x-intercept for both lines. This ensures that winner and loser behavior is evaluated under identical levels of primary competitiveness, making differences in slopes directly interpretable rather than artifacts of differing electoral contexts.



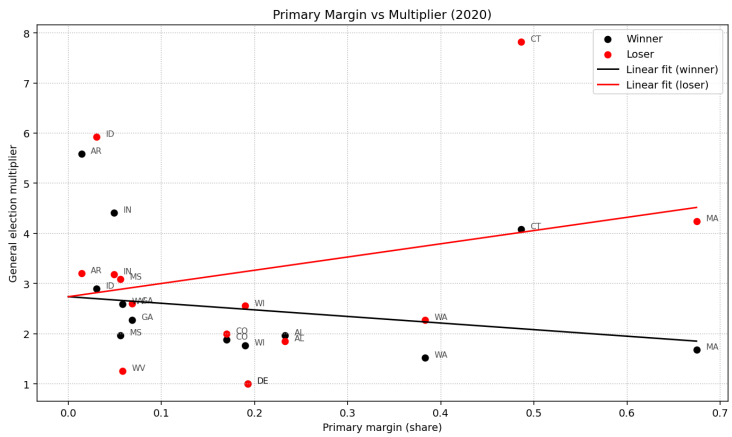
For the 2008 election cycle, the fitted lines exhibit a clear asymmetry between winners and losers. The loser multiplier has a strong positive slope, $\beta_L = 3.816$, while the winner multiplier has a slightly negative slope, $\beta_W = -0.380$. Both lines share the same intercept at approximately 1.893.

This pattern is consistent with a voter complacency interpretation. When the primary is close, both parties expand their turnout by a similar factor in the general election. As the primary becomes more lopsided, however, the losing party's electorate expands more rapidly, suggesting ambition to catch up, while the winning party's turnout growth flattens or even declines. In this sense, increased primary dominance appears to dampen general-election turnout growth for the advantaged party, while activating it for the disadvantaged one.



The 2016 cycle shows a weaker version of the same qualitative pattern. The fitted intercept is approximately 1.713. The loser slope remains positive ($\beta_L \approx 1.523$), indicating that loser multipliers continue to increase as the primary margin grows. In contrast, the winner slope is close to zero ($\beta_W \approx 0.045$), implying that winner turnout expansion is largely insensitive to primary competitiveness.

Compared to 2008, the separation between winner and loser behavior is less clear, but the directional asymmetry remains: losers respond more strongly to lopsided primaries than winners do. Rather than showing clear decline, winner multipliers in 2016 appear roughly flat, which is still consistent with a weak form of complacency.



For 2020, the divergence between winner and loser multipliers is the strongest among the three cycles. The fitted intercept is higher overall ($a \approx 2.739$). The winner slope is sharply negative ($\beta_W \approx -1.313$), while the loser slope is strongly positive ($\beta_L \approx 2.637$).

This creates a clear split: as the primary margin increases, winner multipliers decline while loser multipliers rise. Substantively, this is the most visually striking pattern consistent with voter complacency in the dataset. However, these results must be interpreted cautiously. The 2020 cycle includes many uncontested or lightly contested primaries, which can produce very small primary denominators and mechanically inflate multipliers. In addition, data coverage in 2020 is more uneven due to ETL constraints. A closer inspection of state-level contributions is therefore necessary to determine how much of this pattern is driven by a small number of influential observations.

Overall, the general-election multiplier model reveals a consistent directional asymmetry across election cycles: as primary competitiveness decreases, loser turnout tends to expand more rapidly in the general election, while winner turnout expansion remains flat or declines. This pattern is strongest in 2008 and 2020 and weaker but still present in 2016. While the model is intentionally simple and sensitive to small-denominator effects, the repeated separation between winner and loser slopes provides descriptive evidence consistent with voter complacency operating on the advantaged side of the primary.

5 Shannon Entropy (Primary Competitiveness)

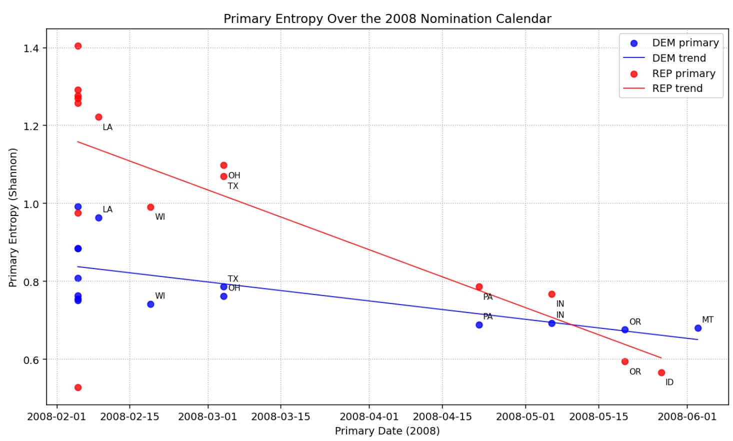
To quantify how competitive (or uncertain) a party's primary election is, we compute Shannon entropy from the distribution of votes across candidates. The intuition is simple: a primary in which votes are split across many candidates reflects higher uncertainty than one dominated by a clear front-runner.

For a given state and party, let $p_1, \dots, p_k$ denote the vote shares of the k candidates in that party's primary, computed by aggregating precinct- or county-level results up to the state level. The Shannon entropy is defined as

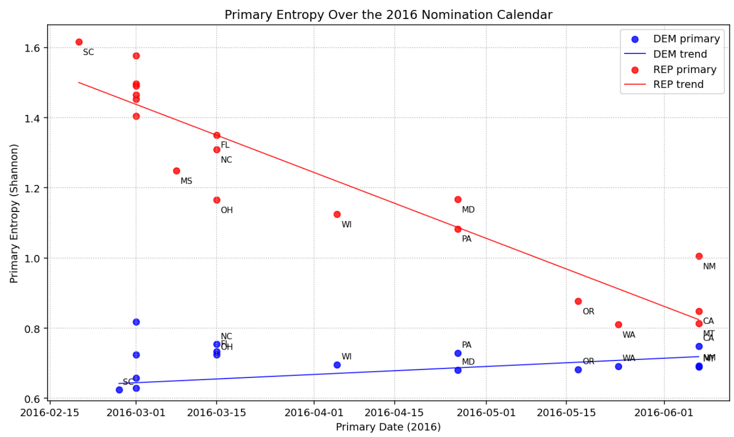
$$H = -\sum_{i=1}^{k} p_i \log(p_i)$$

where p_i is the vote share of candidate i. Entropy is close to zero when one candidate dominates the field, and it increases as vote shares become more evenly distributed across candidates.

For each election cycle, we compute Shannon entropy separately for Democratic and Republican primaries and plot entropy as a function of time throughout the primary season. Election dates are hard-coded using public election calendars, allowing us to align state-level entropy values along a common time axis.

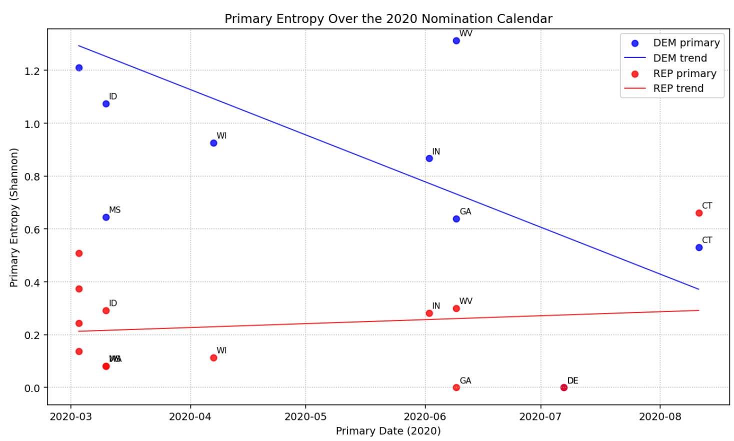


For the 2008 cycle, both Democratic and Republican primary entropy show a clear negative slope over time. This pattern is expected: as the primary season progresses, weaker candidates exit the race, vote shares concentrate around the leading candidates, and overall uncertainty declines. In this sense, the entropy measure behaves as intended, capturing the gradual resolution of competition within each party's primary.



In 2016, the Republican primary continues to show a declining entropy pattern, consistent with the field narrowing over time. The Democratic primary, however, displays a different trend: entropy increases slightly as the primary season progresses. This suggests that, at least in the unweighted state-level aggregation used here, Democratic primary vote shares became more evenly distributed over time rather than more concentrated.

There are several possible explanations for this behavior. First, the observed slope may not be statistically meaningful. A formal regression-based test would be required to determine whether the trend differs significantly from zero. Second, the current entropy calculation weights all states equally, regardless of electorate size. If later-voting states with larger or more competitive contests enter the sample, they may disproportionately influence the aggregate pattern. A natural extension would be to compute a turnout-weighted entropy measure to assess whether this upward trend persists after accounting for state size.



The 2020 cycle shows yet another contrast between parties. Democratic primary entropy declines sharply over time, indicating a rapid consolidation of support as the field narrowed. In contrast, Republican primary entropy exhibits a slight positive slope. This result should be interpreted with caution. The 2020 Republican primary was shaped by unusual circumstances, including the presence of an incumbent president and limited meaningful competition in many states. In addition, data availability and reporting consistency in 2020 posed challenges during the ETL process, and the reduced number of usable states may amplify noise in the entropy trend.

As with 2016, these patterns may be sensitive to weighting choices, state coverage, and timing effects. Expanding the set of states or applying turnout-based weights could help clarify whether the observed increase in Republican entropy reflects a real phenomenon or an artifact of data limitations.

Overall, the entropy analysis illustrates how primary competitiveness evolves over time and varies substantially across election cycles and parties. In cycles like 2008, entropy behaves in a theoretically expected manner, steadily declining as uncertainty resolves. In later cycles, especially 2016 and 2020, the patterns are more mixed, highlighting the importance of institutional context, weighting choices, and data coverage when using entropy as a measure of electoral uncertainty. While Shannon entropy provides a useful summary statistic for primary competitiveness, these results suggest it should be interpreted as one component of a broader analysis rather than as a standalone indicator.

6 Conclusion & Future Directions

This project examines whether patterns consistent with voter complacency appear within U.S. presidential election cycles by linking primary-election dynamics to general-election behavior. Using county-level data from the 2008, 2016, and 2020 elections, we combine descriptive exploration, a simple turnout-based multiplier model, and a time-series entropy analysis to study how primary competitiveness relates to general-election participation.

Across all three cycles, the exploratory vote-share analysis reveals a recurring pattern: stronger Republican primary performance is associated with smaller gains, or even losses, in Republican general-election vote share. While this relationship is purely descriptive, it motivates a closer examination of turnout dynamics rather than vote shares alone. The general-election multiplier model provides that lens. By comparing how turnout expands from the primary to the general election for primary winners and losers, the model consistently shows an asymmetric response: as primaries become more lopsided, loser turnout tends to expand more rapidly, while winner turnout expansion flattens or declines. This pattern is strongest in 2008 and 2020 and weaker, but still present, in 2016. Taken together, these results provide descriptive evidence consistent with voter complacency operating on the advantaged side of the primary.

The Shannon entropy analysis complements this picture by tracking how primary uncertainty evolves over time. In 2008, entropy behaves as expected, declining steadily as candidates exit and competition resolves. In 2016 and 2020, entropy patterns are more mixed and sensitive to aggregation and weighting choices, underscoring the role of institutional context and data coverage. Rather than serving as direct evidence of complacency, entropy functions here as a contextual measure of uncertainty that helps interpret the primary environment in which turnout decisions are made.

Several limitations are worth emphasizing. First, all analyses in this project are descriptive. They do not identify causal effects of primary competitiveness on general-election turnout. Second, the general-election multiplier is sensitive to small primary denominators, particularly in uncontested or lightly contested races, which are more common in recent cycles. Third, data availability and reporting heterogeneity, especially in 2020, limit state coverage and may amplify the influence of a small number of observations. As a result, the findings should be interpreted as suggestive patterns rather than definitive conclusions.

There are several natural directions for future work. One extension would be to incorporate turnout-weighted entropy and turnout-weighted multiplier regressions to reduce the influence of small states and uncontested primaries. Another would be to formally test slope differences using regression-based inference or hierarchical models that account for state-level heterogeneity. Expanding the analysis to include earlier election cycles, or extending the framework symmetrically to Democratic overperformance, would also help assess the robustness and generality of the observed patterns. Finally, continued refinement of the ETL data-cleaning pipeline would improve data quality, particularly for the current results of the 2020 election cycle.

Overall, this project demonstrates that even simple, transparent models can reveal systematic differences in how advantaged and disadvantaged parties translate primary participation into general-election turnout. While voter complacency cannot be directly observed, the repeated asymmetry between primary winners and losers across multiple election cycles suggests that primary competitiveness plays an important role in shaping general-election engagement.